

WOOD CUTTING PROCESS REPORT

Process #1

PROCESS INFORMATION

Process ID	1
Start Time	21/06/2026 00:59:01
End Time	21/06/2026 01:59:01
Duration	01:00:00
Wood Type	Native beech
Category	Hardwood

PRODUCTION SUMMARY

Product Width : 30 mm
Product Height : 200 mm
Production Volume : 0.850 m³
Total Cost : 1020.00

MACHINE CONDITIONS

Cutting Speed	64.00 m/min
Feed Rate	11.00 m/min
Shaft Speed	3100 RPM
Consumed Electricity	8.40 kWh

CRITICAL CUTTING ANALYSIS

Cutting Force	53.49 N
Active Force	54.23 N
Friction Force	26.31 N
Thrust Force	8.91 N
Shear Force	34.57 N
Normal Force To Shear	40.81 N
Normal Force To Rake	46.57 N
Cutting Moment	14.55 N.m

PERFORMANCE INDICATORS

Duration 01:00:00	Production Rate 0.850 m³/hr	Energy Efficiency 0.101 m³/kWh
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